J1

+3V3 1
GND 2
GPIO15 3
GPIO2 4
DRV_EN 5
TMC_TX 6
TMC_UART 7
X_STEP 8
X_DIR 9
Y_STEP 10
Y_DIR 11
12
13
DIAG_Y 14
LED_Y 15

HW-394 3V3 row (power, board edge side)
ESP32 DevKit, 3V3 column

J2

1
GND 2
DIAG_X 3
4
GPIO14 5
GPIO27 6
GPIO26 7
GPIO25 8
JOY_SW 9
LED_X 10
JOY_VRY 11
JOY_VRX 12
GPIO39 13
GPIO36 14
15

HW-394 VIN row (opposite, module side)
ESP32 DevKit, VIN column

J3

DRV_EN 1
X_MS1 2
X_MS2 3
TMC_UART 4
5
6
X_STEP 7
X_DIR 8

TMC2209 X logic

J5

VM 1
GND 2
X_A2 3
X_A1 4
X_B1 5
X_B2 6
+3V3 7
GND 8

TMC2209 X power

J7

X_DIAGP1 1
X_DIAGP2 2

TMC2209 X DIAG/INDEX

JP1

GND 1
2
3 +3V3

X MS1

JP2

GND 1
2
3 +3V3

X MS2

JP5

X_DIAGP1 1
2
3 X_DIAGP2

X_DIAGSEL

X DIAG sel

R2

X_DIAGP1
X_DIAG

1k
DIAG series

C1

VM
GND

100u 35V

J9

X_A2 1
X_A1 2
X_B1 3
X_B2 4

MOTOR X

D2

GND
X_LED_A

blue
X motor active

J4

DRV_EN 1
Y_MS1 2
Y_MS2 3
TMC_UART 4
5
6
Y_STEP 7
Y_DIR 8

TMC2209 Y logic

J6

VM 1
GND 2
Y_A2 3
Y_A1 4
Y_B1 5
Y_B2 6
+3V3 7
GND 8

TMC2209 Y power

J8

Y_DIAGP1 1
Y_DIAGP2 2

TMC2209 Y DIAG/INDEX

JP3

GND 1
2
3 +3V3

Y MS1

JP4

GND 1
2
3 +3V3

Y MS2

JP6

Y_DIAGP1 1
2
3 Y_DIAGP2

Y_DIAGSEL

Y DIAG sel

R3

Y_DIAGP1
Y_DIAG

1k
DIAG series

C2

VM
GND

100u 35V

J10

Y_A2 1
Y_A1 2
Y_B1 3
Y_B2 4

MOTOR Y

D3

GND
Y_LED_A

blue
Y motor active

R5

LED_Y
Y_LED_A

100
LED series

R6

LED_X
X_LED_A

100
LED series

R1

TMC_TX
TMC_UART

1k
UART TX

R4

DRV_EN
+3V3

2.2k
EN pull-up

J11

VM_IN 1
GND 2

12V IN

F1

VM
VM_IN

3A polyfuse

D1

VM
VM_F

SR340
cathode pin 1 to VM

C3

VM
GND

470u 25V
input bulk

J13

GND 1
+3V3 2
JOY_VRX 3
JOY_VRY 4
JOY_SW 5

JOYSTICK
pin header, dupont leads to the joystick module

C6

JOY_VRX
GND

100n

C7

JOY_VRY
GND

100n

J14

GPIO15 1
GPIO2 2
GPIO14 3
GPIO27 4
GPIO26 5
GPIO25 6
GPIO39 7
GPIO36 8
+3V3 9
+3V3 10
+3V3 11
GND 12
GND 13
GND 14

spare GPIO + power pads
bare pads, solder a header as needed

H1

M3

H2

M3

H3

M3

H4

M3

ESP32 HW-394 DevKit + 2x BTT TMC2209, generated by hardware/pcb/generate.py

hyperscope

Sheet: /
File: hyperscope-stage.kicad_sch

Title: hyperscope stage carrier

Size: A2	Date:	**Rev: C**
KiCad E.D.A. 10.0.6		Id: 1/1